

DINAKAR BALEHONNIGA LINGARAJU

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EXPERIENCE

Arizona State University, SAILS

Testing Accommodations | Tempe, AZ | Oct 2025 – May 2026

- Identified and automated a manual Excel record-update workflow using Python, processing 2,000+ student records in under 10 seconds, eliminating human error, and freeing staff hours weekly for higher-value work.

Amadeus Labs, Bangalore, India

Software Development Engineer | Aug 2022 – Jul 2024

- Cut page load times by nearly 50% by migrating internal and customer-facing web applications to Angular 14, improving responsiveness and frontend maintainability.
- Built Python, C++, and SQL backend modules for real-time data services while resolving frontend/backend defects to improve application stability and reduce recurring issues.
- Mentored interns, supported sprint planning and documentation, and presented an award-nominated project to global stakeholders during standups and quarterly reviews.

Software Engineer Intern | Feb 2022 – Jul 2022

- Reduced recurring ad hoc reporting requests by building a self-service reporting app with a Flask backend and Angular frontend, enabling non-technical users to generate custom reports without writing SQL.
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PROJECTS

PawPal+ AI Pet Care Advisor | Python, Groq API, LLaMA 3.3, Streamlit, LLM Integration | [Live Demo](#) | [github](#)

- Built a production-grade AI pet care advisor with conversation history, context-aware pet profile prompts, and a confidence scoring system that auto-rejects off-topic queries via guardrails
- Engineered multi-turn conversation architecture injecting pet context into LLM history, enabling contextually aware follow-up answers without repeating prior advice
- Achieved 15/15 evaluation harness tests across 5 categories (Nutrition, Exercise, Health, Edge Cases, Guardrails) with 89% avg confidence on legitimate inputs and 100% guardrail trigger rate.

AgentOS | Python, FastAPI, Cognee, LLM Integration, React | [github](#)

- Architected a 3-agent pipeline (Planner → Researcher → Writer) backed by Cognee's graph-vector memory layer, enabling knowledge to accumulate and improve across.
- Engineered semaphore-based parallel researchers, PDF/URL ingestion, domain guardrails, and real-time agent event streaming over.

Medication Quality Analytics | Python, DuckDB, SQL, pandas, Power BI | [github](#)

- Built an end-to-end healthcare analytics pipeline on public FDA NDC and CMS data, ingesting, cleaning, and validating 2,988 medication products and 5,432 hospital records.
 - Modeled a DuckDB analytical warehouse with SQL reporting views; surfaced 250 standards requiring review and measured 70.8% RxNorm completeness across 3 data domains.
 - Delivered a 4-page Power BI dashboard covering medication KPIs, hospital quality benchmarks, data quality monitoring, and prioritized improvement opportunities.
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SKILLS

Languages: Python, SQL, C++

Web Technologies: React, Angular, Flask, HTML5, CSS3, JavaScript, REST APIs

AI/ML: LLM Integration, Prompt Engineering, Guardrails, Evaluation Harnesses, scikit-learn, pandas, NumPy

Data & Analytics: DuckDB, MySQL, Power BI, Tableau, Streamlit

Tools: Git/GitHub, Groq API, FastAPI MS Office, VS code, Cursor, GitHub Copilot, Claude Code

EDUCATION

- Arizona State University**, Tempe, AZ | M.S. Data Science | GPA: 3.97 | 08/24 – 05/26
 - PES College of Engineering**, Mandya, India | B.E. Computer Science | CGPA: 8.77 | 08/18 – 07/22
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CERTIFICATIONS

- Foundations of AI Engineering** – CodePath | Feb 2026 – Apr 2026
- Intro to Web Development** – CodePath | Feb 2026 – Apr 2026
- Applications of AI Engineering** – CodePath (ongoing)